

INDUSTRY Project Report

ON

Integratex



Submitted To

Master of Computer Application (MCA)

LDRP Institute of Technology & Research

Gandhinagar BY

SNEHA KALATHIYA (23MCA30017)

URVISHA LATHIYA (23MCA30021)

Under the Guidance of

Prof. Riya Patel

LDRP Institute of Technology and Research

MCA Department

April2025

LDRP Institute of Technology & Research

Sector – 15, Kh-5 Circle, Gandhinagar-382015.

CERTIFICATE

This is to certify that the project titled **Integratex** is the bona fide work carried out by **MS. SNEHA KALATHIYA (23MCA30017), MS. URVISHA LATHIYA (23MCA30021)** a students of LDRP Institute of Technology and Research MCA Department, during the academic year 2024-25, in partial fulfillment of the requirements for the award of the degree of MCA Semester 4.

Date:

Prof. Riya Patel

Prof. Mahendra Khambhalia

Internal Guide

Head of Department

Completion Certificate



Date: 1-April-2025

Certificate of Internship

Dear Sneha and Urvisha,

This is to certify that Kalathiya Sneha Kishorbhai (23MCA30017) , Lathiya Urvisha Rameshbhai (23MCA30021) has successfully completed her internship phase 1 with EduNeuro Solutions LLP from 10 Oct 2024 to 31 Mar 2025 . During her tenure, she played a pivotal role in the development of an IntegrateX, the skill gap between students and the industry by providing a structured system for skill assessment and recruitment.

Throughout the internship, Sneha and Urvisha demonstrated exceptional technical proficiency, analytical thinking, and a proactive approach to problem-solving. Her contributions significantly enhanced the platform's functionality, making it a valuable resource for educators nationwide. Her dedication, creativity, and commitment to delivering high-quality solutions were evident in her work.

We acknowledge and appreciate her outstanding efforts and wish her continued success in all her future endeavors.

Sincerely,

Dhruv Saidava
Director
EduNeuro

ABSTRACT

The **IntegrateX Platform** aims to bridge the skill gap between **students and the industry** by providing a structured system for skill assessment and recruitment. It replaces traditional resumes with dynamic, **validated skill profiles** that include projects, certifications, and real-world contributions.

Key Features:

- **Skill-Tracking System:** Structured, branch-specific skill checklists for students.
- **Validation Mechanisms:** Integration with GitHub, LinkedIn, and certification platforms.
- **Recruitment Efficiency:** A streamlined process for recruiters to filter and compare candidates based on skills.
- **Academia-Industry Collaboration:** Professors and placement cells can monitor student progress and tailor curricula to industry needs.

Through advanced **data analytics, seamless integrations, and a user-friendly experience**, this platform transforms how students prepare for careers and how companies find the right talent.

ACKNOWLEDGEMENT

With profound pleasure and deep gratitude, we extend our heartfelt thanks to:

- Our **H.O.D, Prof. Mahendra Khambhalia**, and the MCA Department of LDRP-ITR, for their guidance and the opportunity to explore this project.
- Our mentor **Prof. Riya Patel**, for valuable insights, technical guidance, and continuous support throughout the development process.
- Our **Team Leader, Mr. Dhruv Saidava**, for his strategic direction and constant motivation.
- Our peers and family members, whose encouragement and constructive feedback have been instrumental in shaping this project.

INDEX

1. INTRODUCTION.....	1
o Purpose of the Document.....	2
o Scope of the Platform.....	3
o Objectives and Goals.....	4
2. PROBLEM DEFINITION.....	5
o Challenges in Skill Assessment & Hiring.....	5
o Limitations of Existing Systems.....	6
3. SYSTEM REQUIREMENT SPECIFICATION.....	7
o Current System vs. Proposed System.....	7
o Functional Requirements.....	7
o Non-Functional Requirements.....	8
o System Modules Overview.....	8
4. TOOLS & TECHNOLOGY.....	9
5. Table Structure.....	10
o Data Dictionary.....	19
o User Interface Overview.....	13
6. PROJECT ANALYSIS.....	16
o Use Case Diagrams.....	16
o System Flow Diagrams.....	17
o E-R Diagrams.....	18
o Data Flow Diagrams (DFD).....	19
7. DESIGN.....	24
8. TESTING.....	39
o White Box Testing.....	39
o Unit Testing & Integration Testing.....	39
o Security & Performance Testing.....	39
9. STRENGTH OF PROJECT.....	40
o Innovation & Impact on Industry.....	40
o Scalability & Future Enhancements.....	40
10. FUTURE ENHANCEMENTS.....	41
11. USER MANUAL & TECHNICAL DOCUMENTATION.....	42
o Step-by-step Platform Guide.....	42
o Admin & User Functions.....	43
12. Conclusion.....	44
13. REFERENCES.....	45

INTRODUCTION

1. Overview

India produces over 1.5 million engineering graduates annually, yet a significant skill gap exists between academia and industry requirements. Traditional resumes and grades often fail to accurately reflect a student's real-world capabilities. As a result, many graduates struggle to secure employment despite their academic achievements. IntegrateX is designed to bridge this gap by providing a structured, skill-based assessment and validation platform that aligns student competencies with industry expectations.

2. Purpose

The primary purpose of IntegrateX is to replace outdated recruitment and assessment methods with a dynamic, skill-based profiling system. By integrating skill tracking, validation mechanisms, and industry-specific requirements, the platform aims to enhance employability and streamline the hiring process for recruiters. It also serves as a structured learning path for students, helping them develop in-demand skills from the beginning of their academic journey.

3. Key Challenges in the Current System

1. Mismatch Between Education and Industry Needs:

- A survey by the India Skills Report (2023) found that only 46% of engineering graduates are employable due to a lack of practical, industry-relevant skills.
- Colleges often focus on theoretical knowledge rather than practical skill development.

2. Inefficiencies in Recruitment:

- Recruiters rely on generic resumes and GitHub profiles, making it difficult to assess a candidate's practical expertise.
- Initial screening processes are time-consuming and ineffective.

3. Fragmented Academic Skill Development:

- Professors and placement cells lack structured tools to monitor students' technical growth.
- Students often struggle to identify and develop skills that are in high demand.

4. How IntegrateX Works

IntegrateX functions as a centralized platform that brings together students, professors, placement cells, and recruiters. It features:

- **Dynamic Skill Checklists:** Industry-aligned, branch-specific checklists ensure students focus on relevant skills.
- **Skill Validation:** Students can verify their competencies by linking GitHub projects, uploading certifications, and completing real-world assessments.

- **Recruitment Optimization:** HR professionals can filter, compare, and shortlist candidates based on validated skill data, reducing time-to-hire.
- **Professor & Placement Dashboards:** Faculty members can track student progress, customize skill checklists, and generate placement readiness reports.

5. Expected Impact

- **For Students:** A structured, trackable learning journey that improves employability.
- **For Professors & Placement Cells:** Simplified monitoring of student progress and better alignment with industry needs.
- **For Recruiters:** Efficient candidate screening, leading to higher-quality hires in less time.
- **For the Industry:** A workforce better prepared to meet evolving technological demands.

1.1 PURPOSE OF THE DOCUMENT

The purpose of this document is to establish a comprehensive framework for the **design, development, and implementation** of the IntegrateX Platform. It aims to define the following:

1. **Standardized Skill Validation System**
 - Eliminates reliance on traditional resumes and subjective assessments.
 - Ensures students develop industry-relevant skills through structured tracking and validation.
2. **Efficient Recruitment Process**
 - Provides recruiters with **pre-validated skill profiles** of candidates.
 - Reduces time spent on screening and shortlisting applicants.
3. **Academic-Industry Collaboration**
 - Helps **professors and placement cells** track student progress.
 - Aligns engineering curricula with current market demands.
4. **Stakeholder Alignment**
 - Serves as a **blueprint** for all stakeholders, including students, professors, recruiters, and platform administrators.
 - Ensures smooth collaboration by defining clear **roles, responsibilities, and expectations**.

This document will guide the **development, implementation, and evaluation** of the platform, ensuring it meets the needs of all users effectively.

1.2 SCOPE OF THE PLATFORM

The IntegrateX platform addresses the limitations of conventional **hiring, skill assessment, and academic tracking** by implementing a structured, **data-driven approach** to evaluating students' competencies. The platform's scope extends across multiple stakeholders, offering unique benefits to each.

1.2.1 Student Features

- Maintain **branch-specific skill checklists**.
- Self-assess progress and validate skills through **projects, certifications, and practical applications**.
- Integrate with **GitHub, LinkedIn, and learning platforms** (e.g., Coursera, Udemy).
- Generate a **structured, validated skill profile** for recruiters.

1.2.2 Professor & Placement Cell Features

- **Monitor and manage student progress** using role-specific dashboards.
- Align skill checklists with academic coursework.
- Generate **placement readiness reports** for students.
- Communicate skill gaps and suggest areas of improvement.

1.2.3 Industry Professional & Recruiter Features

- **Search and filter** candidates based on **specific skill requirements**.
- Compare candidate profiles using a **standardized skill checklist**.
- **Shortlist candidates** for interviews directly through the platform.
- Verify student skills through **project showcases, linked certifications, and external integrations**.

1.2.4 Admin Features

- Oversee **user management, security settings, and data privacy policies**.
- Configure and maintain **customized skill checklists** for different engineering branches.
- Generate analytics reports on **platform usage, skill trends, and hiring success rates**.
- Implement **role-based access control (RBAC)** for different users.

The platform is designed to scale across multiple educational institutions and industries, making it **flexible and adaptable** to changing skill demands and recruitment trends.

1.3 OBJECTIVES AND GOALS

1.3.1 Enhancing Skill Assessment

- Move beyond traditional resumes to a **dynamic, skill-based profile system**.

- Enable students to **self-assess and validate** their skillsets.
- Provide a **centralized skill-tracking system** for students, professors, and recruiters.

1.3.2 Improving Recruitment Efficiency

- Implement an **automated filtering and comparison** system for candidates.
- **Reduce recruitment time** by offering standardized, validated skill checklists.
- Ensure recruiters can **easily verify skill proficiency** using linked projects and certifications.

1.3.3 Encouraging Continuous Skill Development

- Provide **first-year students** with structured checklists to encourage learning.
- Offer **AI-powered skill recommendations** based on market trends and industry requirements.
- Ensure students remain **industry-ready** through structured learning paths.

1.3.4 Strengthening Academia-Industry Collaboration

- Enable professors to align **skill checklists with industry standards**.
- Bridge the gap between **academic coursework and job market needs**.
- Create a **shared understanding of required skills** between academia and industry.

1.3.5 Ensuring Platform Scalability and Security

- Design a **highly scalable** architecture that supports **thousands of concurrent users**.
- Implement **role-based access control (RBAC) and encryption** for data protection.
- Ensure compliance with **GDPR, CCPA, and industry security best practices**.

The IntegrateX Platform will revolutionize the way **students prepare for careers, recruiters find talent, and educators track skill development**, ultimately **enhancing employability and strengthening the hiring process**.

2. PROBLEM DEFINITION

The growing gap between academic learning and industry expectations has created significant challenges in **skill assessment and hiring**. Employers struggle to identify job-ready candidates, while students lack the necessary tools to track and showcase their skills effectively.

2.1 Challenges in Skill Assessment & Hiring

2.1.1 Lack of Standardized Skill Measurement

- Traditional **resumes and transcripts fail to reflect real-world skills**.
- No uniform framework exists to evaluate technical expertise across students from different institutions.
- Recruiters struggle to distinguish between candidates with similar academic qualifications.

2.1.2 Inefficiency in the Hiring Process

- Companies spend excessive time screening unqualified candidates.
- **Resume-based hiring leads to inaccurate assessments** of a candidate's practical abilities.
- Recruiters lack access to **real-time validated skill data**, leading to **delays in hiring**.

2.1.3 Limited Real-World Exposure for Students

- Students often lack hands-on experience with industry tools and technologies.
- Few opportunities exist for students to validate their skills through projects and internships.
- The gap between **academic coursework and industry expectations** creates **employability challenges**.

2.1.4 Unstructured Academic-Industry Collaboration

- Professors and placement cells struggle to track student progress effectively.
- Industry professionals **lack a direct channel to influence academic curricula**.
- Institutions lack the tools to **align learning objectives with market needs**.

2.2 Limitations of Existing Systems

2.2.1 Resume-Based Hiring Falls Short

- Recruiters rely on **text-based resumes that lack verification.**
- **No direct link between academic achievements and practical applications.**
- Employers face difficulties in **assessing soft skills and problem-solving abilities.**

2.2.2 Limited Integration with Industry Platforms

- Current academic platforms lack integration with **GitHub, LinkedIn, Coursera, and project repositories.**
- No centralized database exists to track students' **skill development and certifications.**

2.2.3 Ineffective Skill Assessment Methods

- Most hiring assessments are **generalized tests that do not measure hands-on expertise.**
- Students **lack guidance on industry-relevant skills** and the best way to validate them.
- Companies **struggle to find candidates with proven expertise** in emerging technologies.

3. SYSTEM REQUIREMENT SPECIFICATION

3.1 Current System vs. Proposed System

Current System Challenges

- Manual tracking of student progress and skills.
- Lack of **real-time validated** skill assessment.
- Time-consuming and inefficient recruitment process.
- No centralized platform for academia-industry collaboration.

Proposed System Benefits

- **Automated skill tracking and validation** using Laravel and MySQL.
- **Centralized candidate profile database** with real-time updates.
- **Faster recruitment process** with AI-based candidate filtering.
- **Seamless integration** with GitHub, LinkedIn, and certification platforms.

3.2 Functional Requirements

- **User Management:** Role-based access control for students, professors, recruiters, and admins.
- **Skill Tracking System:** Branch-specific skill checklists with validation mechanisms.
- **Recruitment Tools:** AI-powered candidate filtering, profile comparison, and direct messaging.
- **Integration Support:** APIs for GitHub, LinkedIn, and third-party certification platforms.
- **Reporting and Analytics:** Placement readiness reports, recruiter activity tracking, and system logs.

3.3 Non-Functional Requirements

- **Scalability:** Ability to handle thousands of users concurrently.
- **Security:** End-to-end encryption, role-based authentication, and compliance with GDPR & CCPA.
- **Performance:** Optimized MySQL queries for fast data retrieval and Laravel caching mechanisms.
- **Reliability:** 99.9% uptime with scheduled backups and failover mechanisms.
- **Usability:** Intuitive UI/UX designed for ease of use across all user roles.

3.4 System Modules Overview

Student Module

- Registration, authentication, and role-based dashboards.
- Profile management, including skill checklists and certifications.

Skill Management Module

- Student progress tracking with **automated skill validation**.
- Integration with external platforms for real-world skill verification.

Recruitment Module

- Candidate search and AI-powered filtering for recruiters.
- Shortlisting, profile comparison, and direct messaging features.

Admin Module

- Platform-wide settings, user access control, and database management.
- Analytics dashboard for monitoring platform performance and user engagement.

4.TOOLS & TECHNOLOGY

Project Title	IntegrateX
Project Category	Web Development
Project Duration	3 Months
Framework	Laravel
Front End	HTML,CSS,JavaScript, jQuery, Bootstrap
Back End	MySQL
Project Guide	Prof. Riya Patel
Submitted By	Urvisha Lathiya
	Sneha Kalathiya
Submitted To	LDRP-ITR College

5..1 Table Structure

1. Table Name: Student

Column name	Data type	Data size	Constraint	Description
Id	Int	20	Primary Key	Id of student
firstname	varchar	20	Not null	Name of student
lastname	varchar	20	Not null	Last name of student
role	varchar	20	Not null	role of student
age	Varchar	20	Not null	age of student
address	Varchar	10	Not null	address of student
phone	Varchar	50	Not null	phone of student
githubuserid	Varchar	20	Not null	userid of student
email	Varchar	20	Not null	emailof student
password	Varchar	06	Not null	password of student
project	Int	20	Not null	project of student
year	Int	20	Not null	year of student
branch_id	Int	11	Foreign key	branch_id of student
placement_status	Varchar	20	Not null	status of student

2. Table Name: Skill

Column name	Data type	Data size	Constraint	Description
Id	Int	20	Primary Key	Id of skill
name	varchar	255	Not null	name of branch

3. Table Name: Skill_subsets

Column name	Data type	Data size	Constraint	Description
Id	Int	11	Primary Key	Id of skill_subsets
skill_id	varchar	20	Foreign Key	skill_id of student
name	varchar	20	Not null	name of skill_subsets

4. Table Name: Student_skills[Pivot table]

Column name	Data type	Data size	Constraint	Description
Id	Int	11	Primary Key	Id of Student_skills

student_id	Int	11	Foreign Key	id of Student_skills
skill_id	Int	11	Foreign Key	id of skill_id
skill_subset_id	Int	11	Foreign Key	id of skill_subset_id

5. Table Name: Project

Column name	Data type	Data size	Constraint	Description
Id	Int	20	Primary Key	Id of project
student_id	Int	20	Foreign Key	Id of student
project_name	varchar	255	Not null	project_name of student
frontend	Varchar	255	Not null	frontend of project
backend	Varchar	255	Not null	backend of project
database	Varchar	255	Not null	database of project
live_demo	Varchar	255	Not null	live_demo of project
description	Varchar	255	Not null	Description of project

6. Table Name: Placement_officer

Column name	Data type	Data size	Constraint	Description
Id	Int	11	Primary Key	Id of placement_officer
officer_name	Varchar	20	Not null	Name of officer
email	Varchar	20	Not null	email of officer
phone	Int	10	Not null	phone of officer
role	Varchar	20	Not null	role of officer
password	Varchar	10	Not null	password of officer

7. Table Name: Branches

Column name	Data type	Data size	Constraint	Description
Id	Int	11	Primary Key	Id of barnch
branch_name	varchar	20	Not null	name of the branch

8. Table Name: Admin

Column name	Data type	Data size	Constraint	Description
Id	Int	11	Primary Key	Id of Admin
name	Int	11	Not Null	Name of admin
email	Int	11	Not Null	Email of admin

password	Int	11	Not Null	Password of admin
----------	-----	----	----------	-------------------

9. Table Name: Recruiters

Column name	Data type	Data size	Constraint	Description
Id	Int	20	Primary Key	Id of recruiter
name	varchar	20	Not null	Name of recruiter
email	varchar	255	Not null	project_name of student
company	Varchar	255	Not null	name of company
company_website	Varchar	255	Not null	Website of company
Linkedin_profile	Varchar	255	Not null	Linkedin profile of company
industry	Varchar	255	Not null	Industry of company
role	Varchar	255	Not null	Role of recruiter
phone	varchar	10	Not null	Phone no of recruiter
password	varchar	08	Not null	Password of recruiter

10. Table Name: campus_recruitmemts

Column name	Data type	Data size	Constraint	Description
Id	Int	20	Primary Key	Id of project
recruiter_id	Int	20	Foreign Key	recruiter id of recruiter
Officer_id	Int	255	Foreign Key	Officer id placement officer
company_profile	Varchar	255	Not null	Profile of recruiter
Job_role	Varchar	255	Not null	Job role
remuneration	Varchar	255	Not null	Remuneration
location	Varchar	255	Not null	Location of job
requirements	Varchar	255	Not null	Requirements of job
Interview_process	Varchar	255	Not null	Interview_process of job

5.1.2 Data Dictionary

Table name	students
Primary key	student_id
Alias Name	None
Where/How Used	To use student can register their details
Description	To store student id, firstname , lastname, role,age, Githubuserid, emaphone, password, address in the table

Table name	Skills
Primary key	skill_id
Alias Name	None
Where/How Used	To use for student can store the skill name
Description	To store skill id , name in the table

Table name	Skill_subsets
Primary key	id
Alias Name	None
Where/How Used	To use for student can store there subskill name
Description	To store skill id , subset skill id , name in the table

Table name	Student_skills
Primary key	id
Alias Name	None
Where/How Used	To use for student can store student skill id
Description	To store student id , skill id and subset id and Student skill id in the table

Table name	projects
Primary key	id
Alias Name	None
Where/How Used	To use for student store project name , frontend , backend, database , live demo and decription
Description	To store project id , name , frontend , backend, database , live demo and decriptionin the table

Table name	branches
Primary key	id
Alias Name	None
Where/How Used	To use for officer can add the branches
Description	To store officer branch id and name in the table.

Table name	Placement_officers
Primary key	id
Alias Name	None
Where/How Used	To use for officer can store id , name ,email , password and role
Description	To store officer id , name ,email , password and role in the table

Table name	Admin
Primary key	id
Alias Name	None
Where/How Used	To use for admin can store id , name, email and password
Description	To store id,name,email and password.

Table name	Recruiter
-------------------	-----------

Primary key	id
Alias Name	None
Where/How Used	To use for recruiter can add company profile,job role,remuneration, location,requirements,interview process.
Description	To store company profile,job role,remuneration, location,requirements,interview process.

Table name	Campus_recruitments
Primary key	id
Alias Name	None
Where/How Used	To use for Campus_recruitments can add name, company,linkedin_profile,industry,role,phone and password.
Description	To store Campus_recruitments can add name, company,linkedin_profile,industry,role,phone and password.

6 Use Cases Diagrams

6.1.1 Gantt Chart

Gantt Chart for IntegrateX (Dec 2 – April 30)

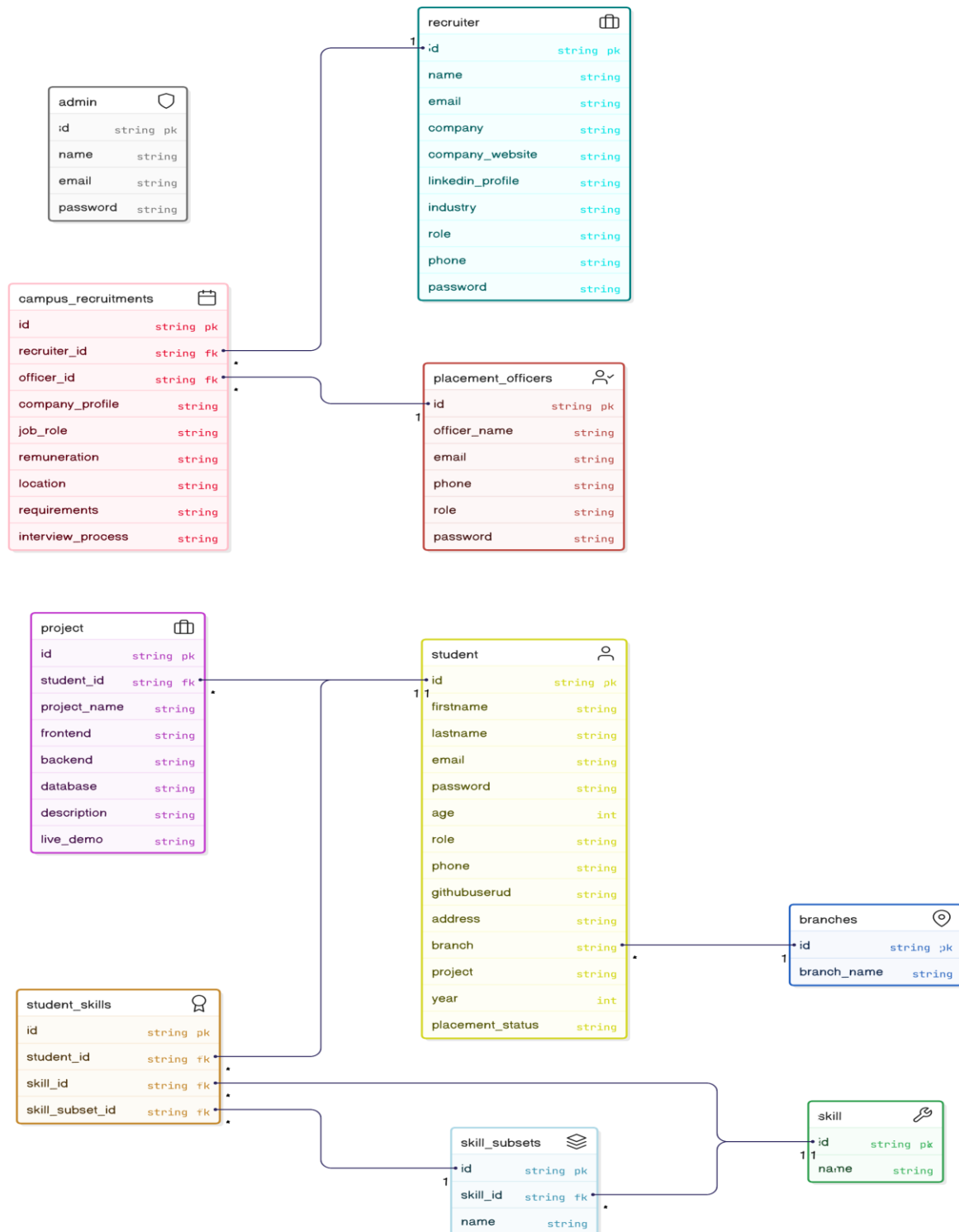
Phase	Tasks	Duration	Start Date	End Date
Phase 1: Student & Professor Modules		Dec – Feb		
Dec 2 – Dec 30	Finalize requirements, wireframes, database setup	4 weeks	Dec 2	Dec 30
Jan 1 – Jan 31	Develop Student Module: Authentication, skill tracking	4 weeks	Jan 1	Jan 31
Feb 1 – Feb 28	Develop Professor Module: Dashboards, reporting	4 weeks	Feb 1	Feb 28
	Testing & feedback on Student & Professor Modules	2 weeks	Feb 15	Feb 28
Phase 2: Industry & Admin Modules		March – April		
March 1 – March 31	Industry Module: Search & filter candidates, shortlisting	4 weeks	March 1	March 31
April 1 – April 30	Admin Module: User management, analytics, system monitoring	4 weeks	April 1	April 30

6.1.2 System Flow

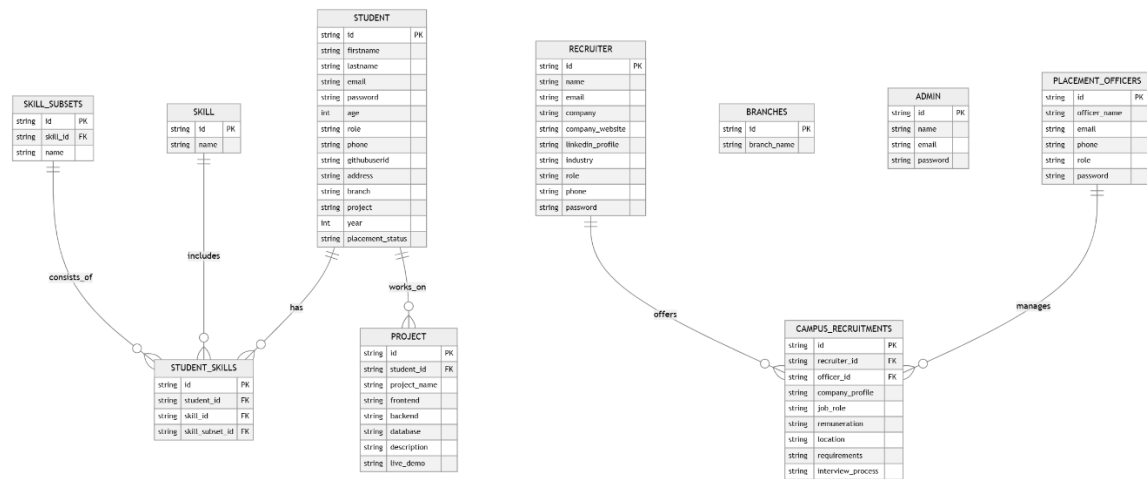
1. **User Login & Authentication**
 - Students, Professors, Placement Officers, Recruiters, and Admins log in.
 - Authentication via college email (Students/Professors) or corporate credentials (Recruiters).
 - Role-based access control (RBAC) determines system permissions.
2. **Student Workflow**
 - Access branch-specific **skill checklist**.
 - Mark **completed skills** and upload validation (certificates, GitHub projects).
 - View **progress tracking dashboard**.
 - Generate **shareable profile link** for recruiters.
3. **Professor & Placement Cell Workflow**
 - View **student progress** via dashboards.
 - Customize **skill checklists** per academic requirements.
 - Generate **placement readiness reports**.
 - Notify students of skill gaps.
4. **Admin Workflow**
 - **Manage users** (add/edit/remove students, professors, recruiters).
 - Configure **skill checklists** per branch.
 - Monitor **platform analytics** (user activity, skills tracked).
 - Oversee **data security & backups**.

6.1.3 ER Diagram

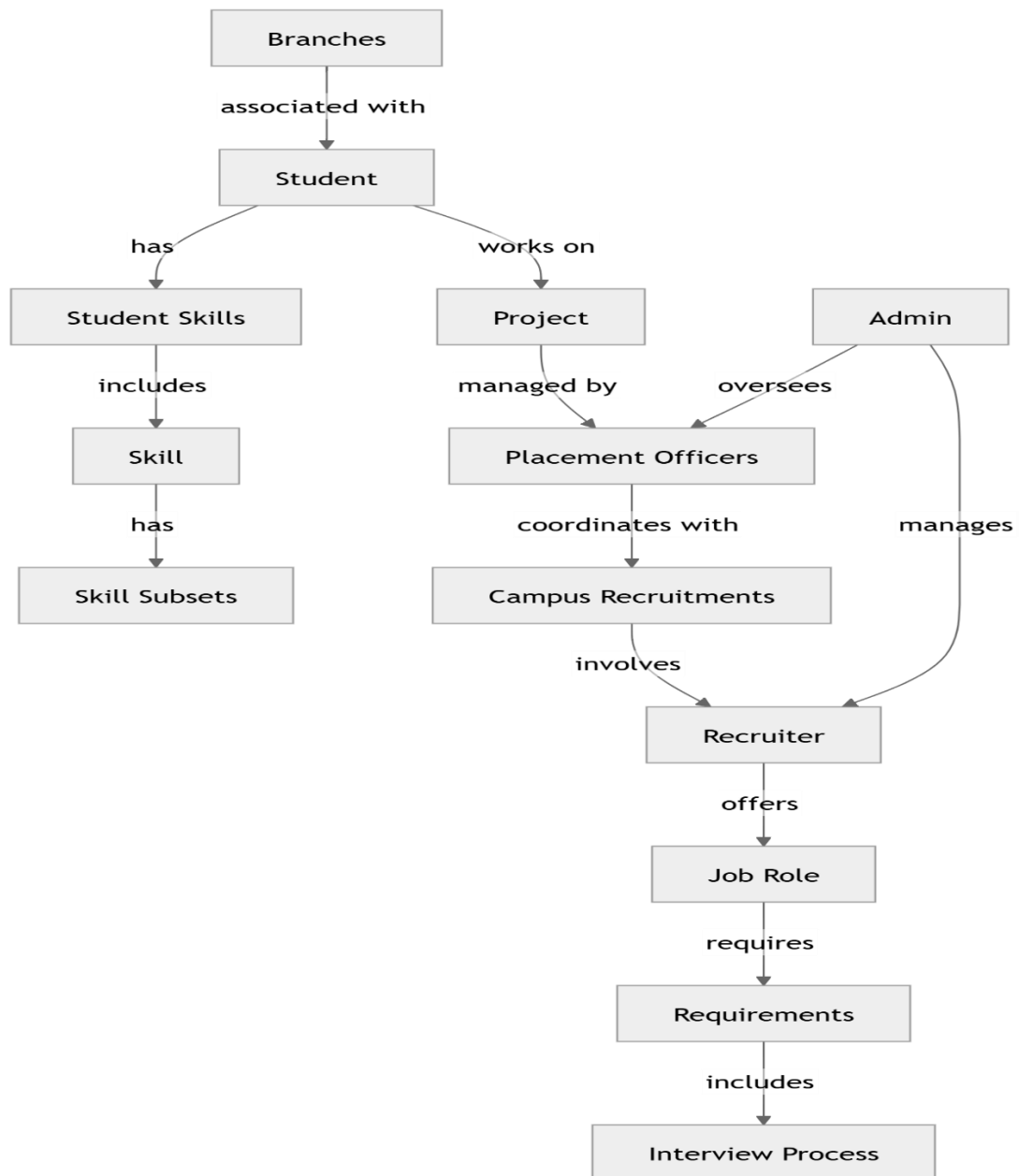
Student Management and Recruitment System



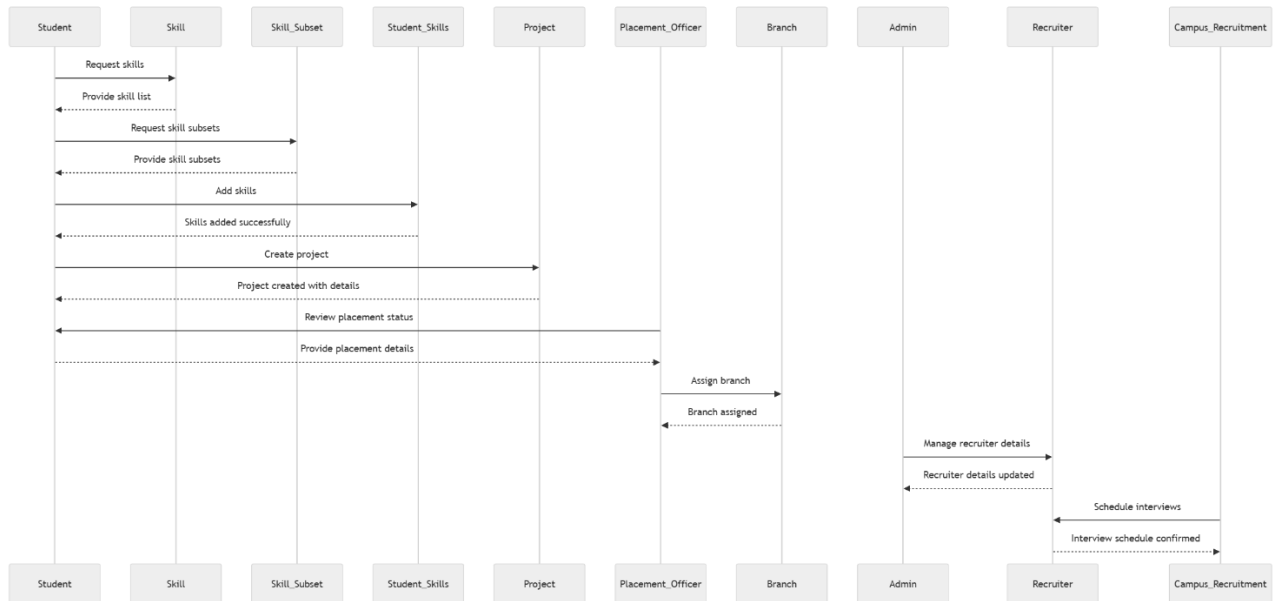
6.1.4 UML (class Diagram)



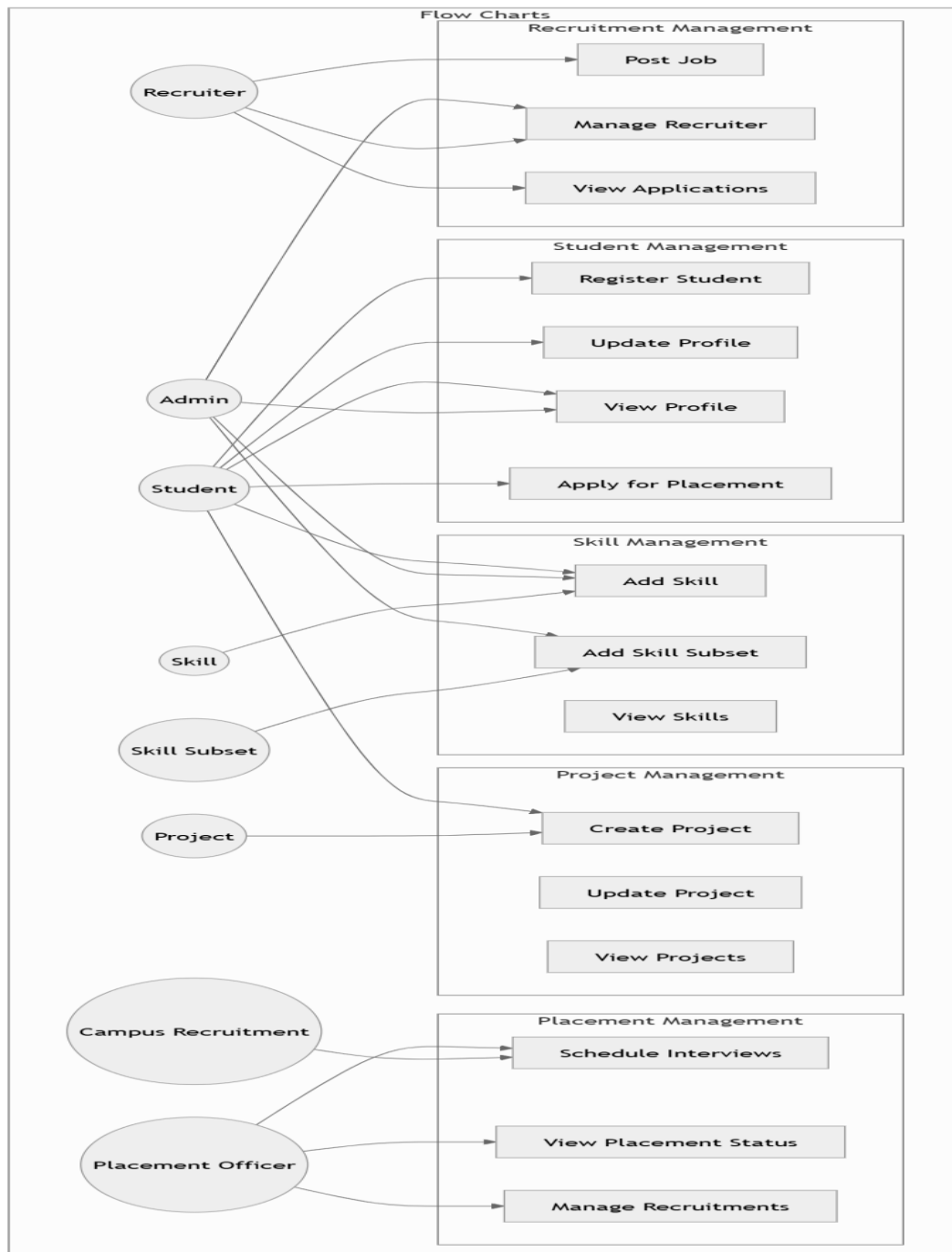
6.1.5 Activity diagram



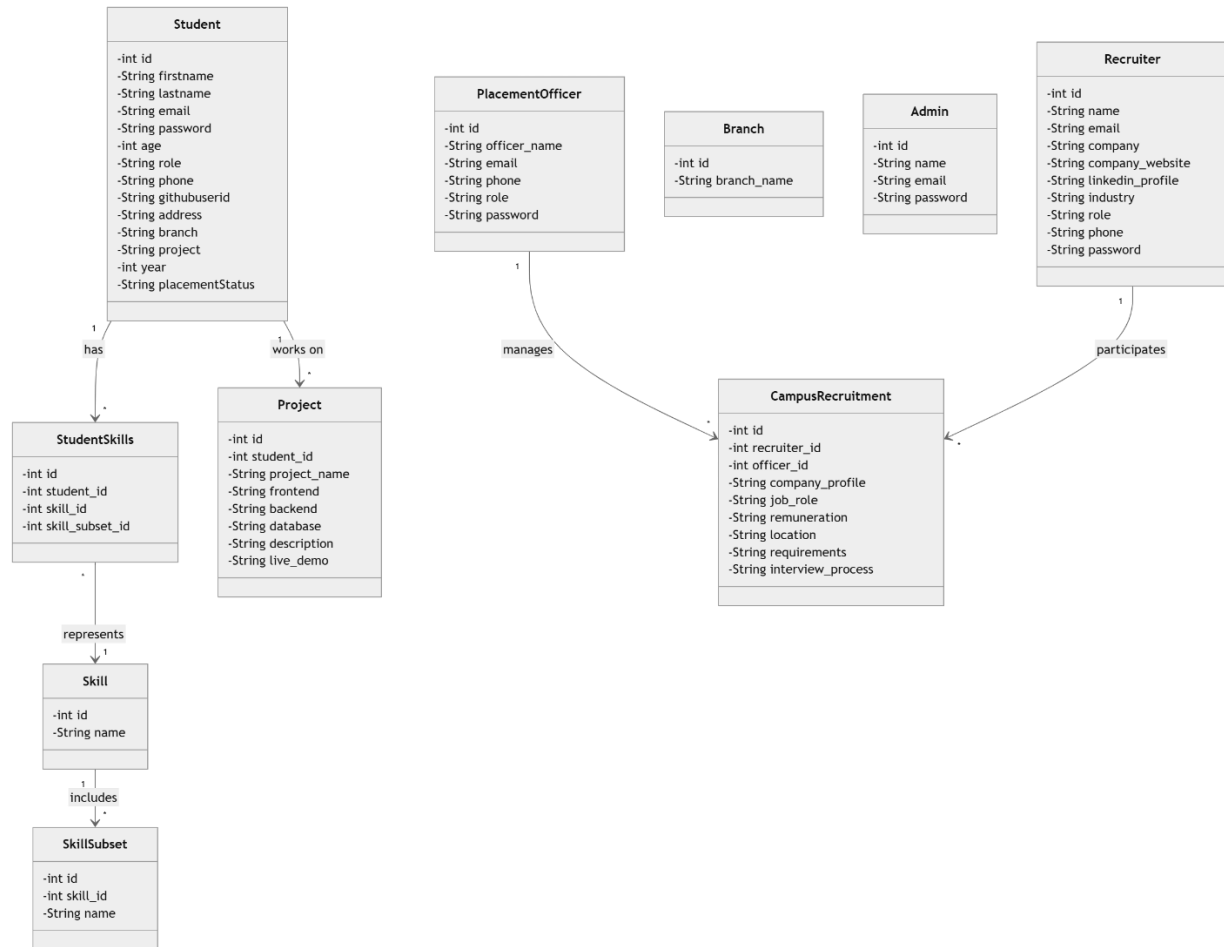
6.1.6 Sequence Diagram



6.1.7 Use Cases

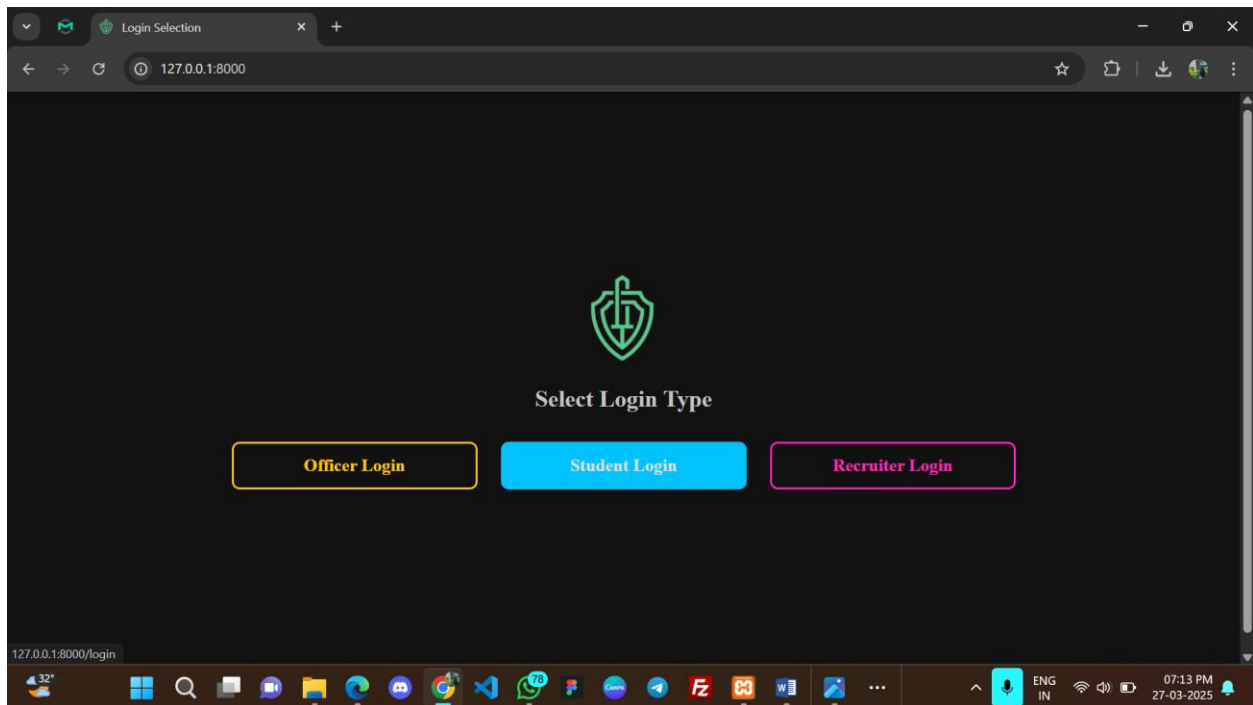


6.1.8 Class diagram

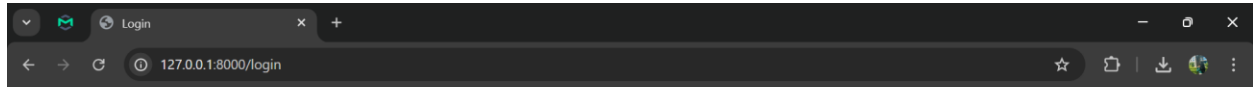


7.Design Report

➔ Student Dashboard



A screenshot of a web browser displaying the 'Register' page. The browser's address bar shows '127.0.0.1:8000/register'. The page has a light gray background and features a white registration form. The form includes the following fields: 'First Name', 'Last Name', 'Email', 'Password', 'Role', 'Age', 'Phone', 'Address', and 'GitHub User ID'. The Windows taskbar at the bottom shows the time as 07:15 PM on 27-03-2025.



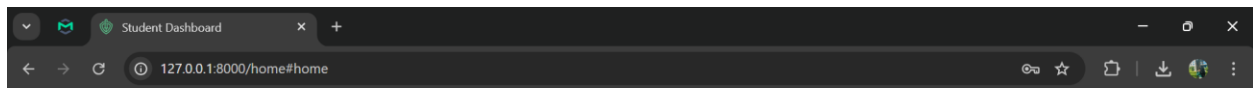
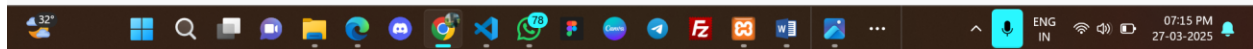
Login

Email

Password

Login

Don't have an account? [Register here](#)





Home

About me

GitHub

Skill

Contact

Projects

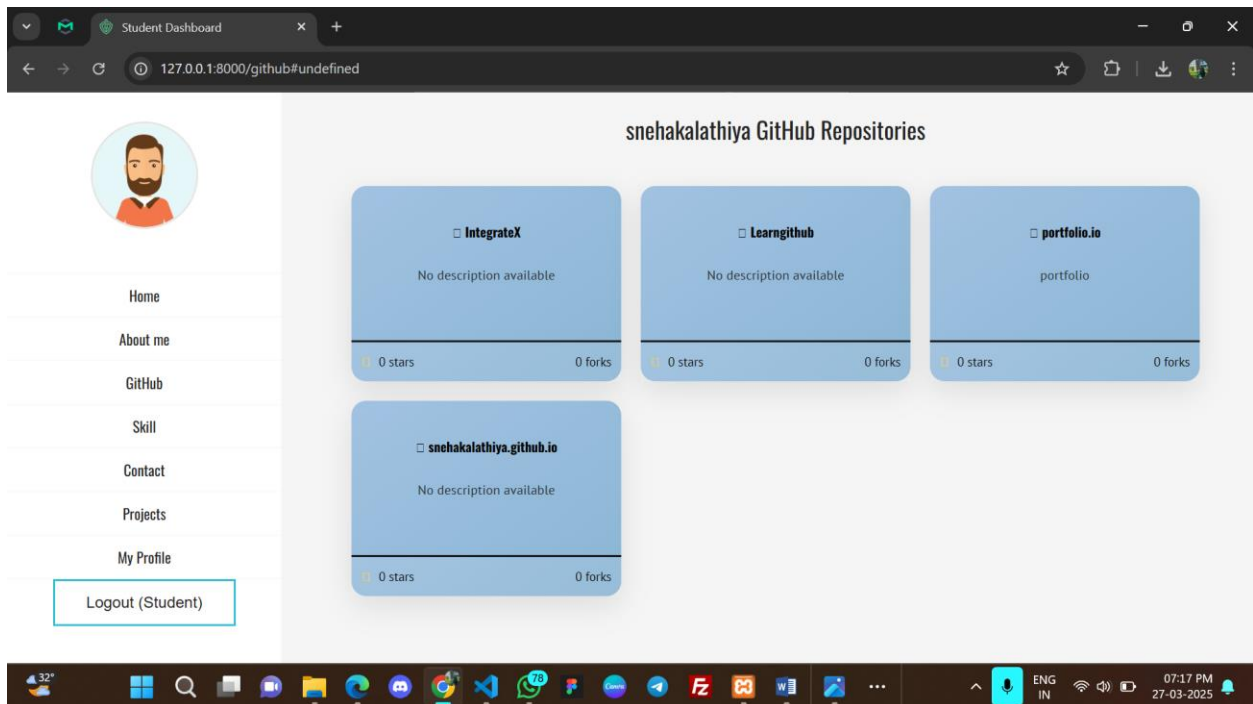
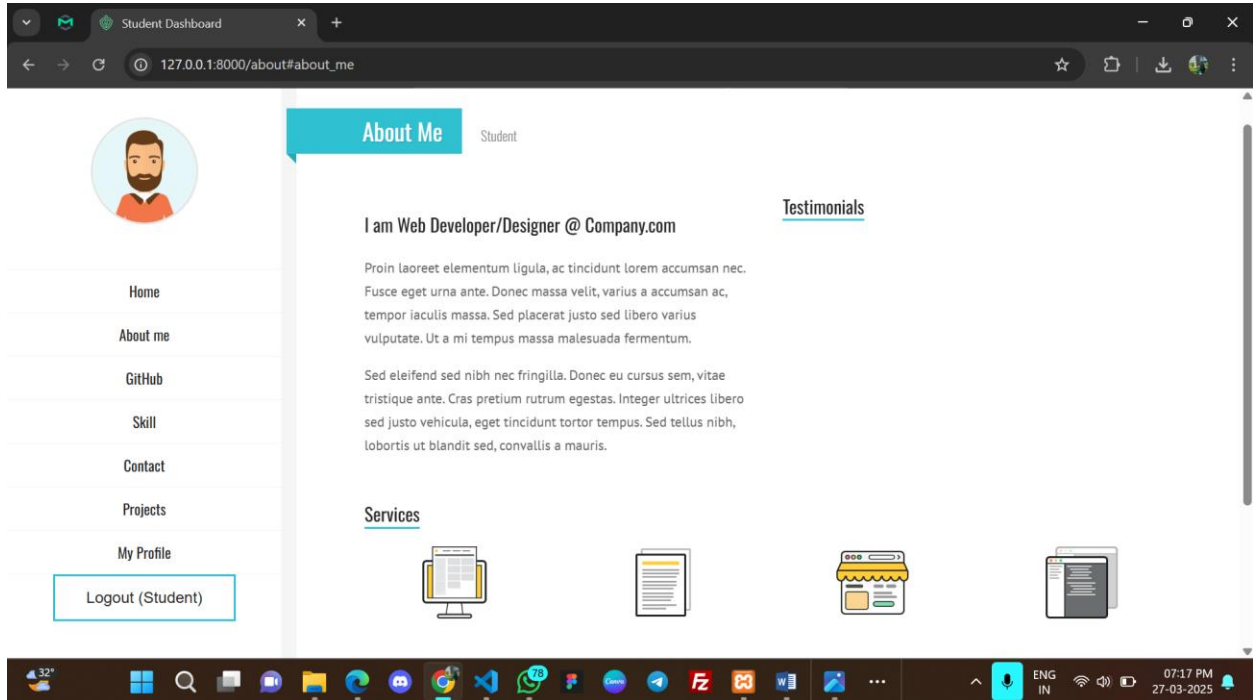
My Profile

Logout (Student)

Welcome, Snehal!

- Role: Student
- Age: 25
- Address: ahmd
- Email: sneha@gmail.com
- Phone: 1231231231
- GitHub Username: snehakalathiya





Student Dashboard

127.0.0.1:8000/skill#undefined



- Home
- About me
- GitHub
- Skill
- Contact
- Projects
- My Profile
- Logout (Student)

Select Your Skills

Programming

- ☒ Select Programming
 - ☒ C
 - ☒ C++
 - ☐ Java
 - ☒ Python
 - ☐ urvisha

Frontend

- ☒ Select Frontend
 - ☒ HTML
 - ☒ CSS
 - ☒ JavaScript
 - ☐ React
 - ☐ Vue

Backend

- ☐ Select Backend
 - ☐ Node.js
 - ☐ Laravel
 - ☐ Django
 - ☐ Spring Boot

DSA

- ☒ Select DSA
 - ☒ Arrays
 - ☒ Linked Lists
 - ☒ Graphs

Version Control

- ☒ Select Version Control
 - ☐ Git
 - ☒ GitHub
 - ☒ GitLab

Snhea

- ☐ Select Snhea
 - ☐ hello

Student Dashboard

127.0.0.1:8000/projects#undefined



- Home
- About me
- GitHub
- Skill
- Contact
- Projects
- My Profile
- Logout (Student)

Live Demo (Optional):

Description:

[Submit](#)

Your Projects

IoT Smart Home

Frontend: React

Backend: Laravel

Database: MySQL

Description: A home automation system using IoT.

Live Demo: [View](#)

AI Chatbot

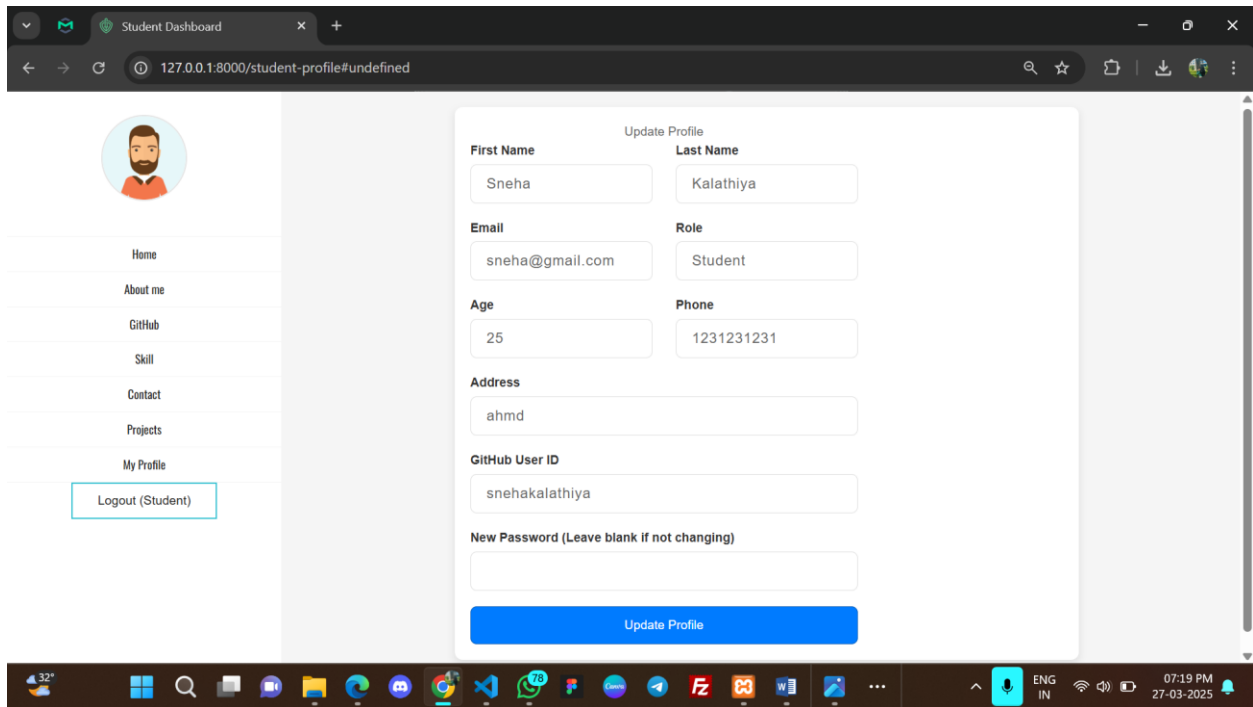
Frontend: Vue.js

Backend: Django

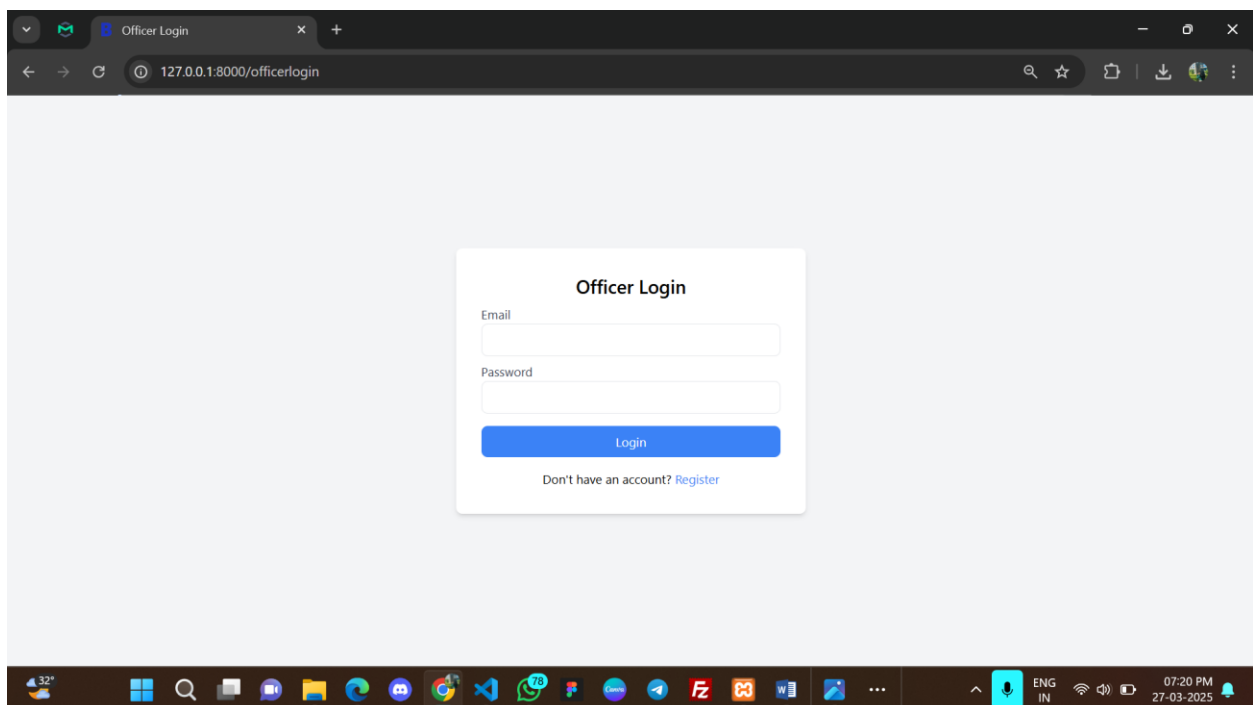
Database: PostgreSQL

Description: An AI chatbot for customer support.

Live Demo: [View](#)



Professor Login



Placement Dashboard

127.0.0.1:8000/index

Dashboard

Show Skill

Reporting

Manage Skills

Branch

Update Profile

Logout (Placement)

Welcome, Sneha

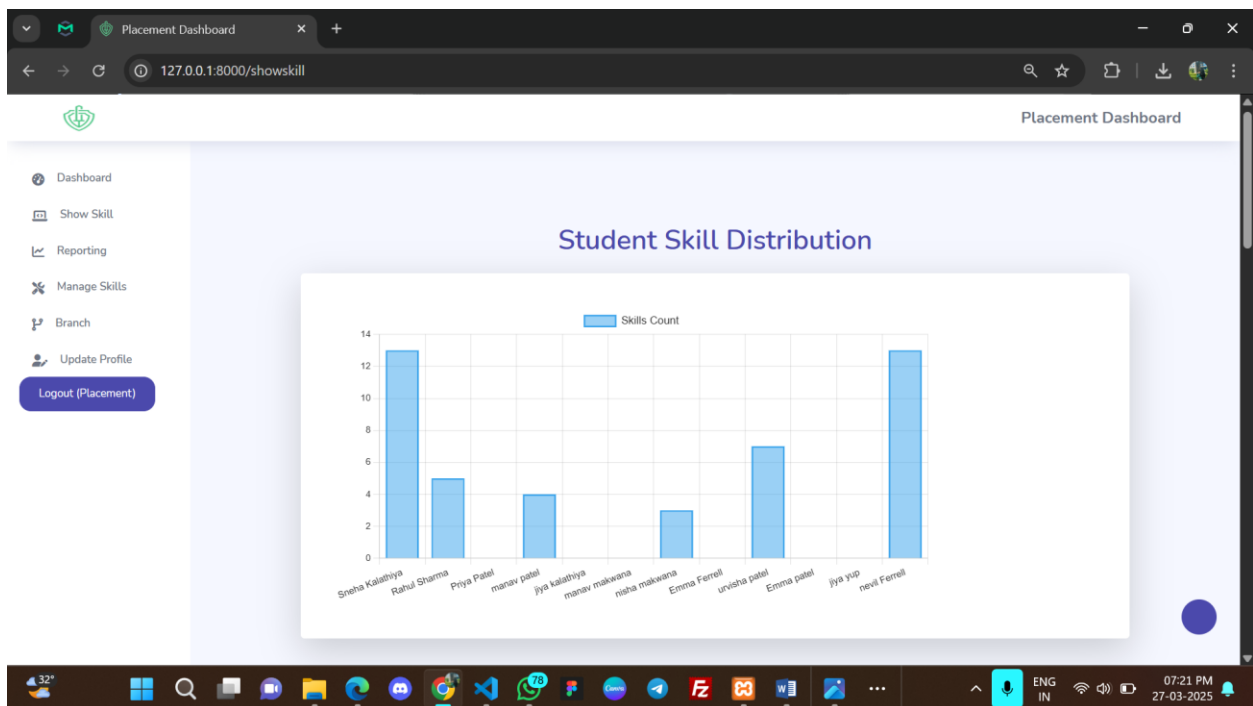
Your Role: Head

Email: sneha@student.com

32°

ENG IN

07:21 PM 27-03-2025



Placement Dashboard

Student Skill Dashboard

Student ID	Student Name	Skills
1	Sneha Kalathiya	- C (Programming) - C++ (Programming) - Python (Programming) - HTML (Frontend) - CSS (Frontend) - JavaScript (Frontend) - Arrays (DSA) - Linked Lists (DSA) - Graphs (DSA) - Trees (DSA) - GitHub (Version Control) - GitLab (Version Control) - hello (hello)
2	Rahul Sharma	- CSS (Frontend) - JavaScript (Frontend) - Linked Lists (DSA) - Graphs (DSA)

Placement Dashboard

Student Placement Dashboard

2024 BCA Filter by Skill Apply Filters

Student ID	Name	Year	Branch	Skills
1	Sneha Kalathiya	2024	BCA	✓ C (Programming) ✓ C++ (Programming) ✓ Python (Programming) ✓ HTML (Frontend) ✓ CSS (Frontend) ✓ JavaScript (Frontend) ✓ Arrays (DSA) ✓ Linked Lists (DSA) ✓ Graphs (DSA) ✓ Trees (DSA) ✓ GitHub (Version Control) ✓ GitLab (Version Control) ✓ hello (hello)
6	manav makwana	2024	BCA	
11	Emma patel	2024	BCA	

Download PDF

Placement Dashboard

127.0.0.1:8000/assign-skill

Placement Dashboard

Show Skill

Reporting

Manage Skills

Branch

Update Profile

Logout (Placement)

Manage Skills & Assign to Students

Add New Skill

Enter Skill Name

Add Skill

Add New Sub-Skill

Select Skill

Enter Sub-skill Name

Add Sub-Skill

Skills & Sub-Skills

Skill

Sub-skills

32°

Windows Taskbar

ENG IN

07:23 PM 27-03-2025

Placement Dashboard

127.0.0.1:8000/assign-skill

Placement Dashboard

Skills & Sub-Skills

Skill

Sub-skills

32°

Windows Taskbar

ENG IN

07:24 PM 27-03-2025

Skill	Sub-skills
Programming	☒ C ☒ C++ ☒ Java ☒ Python ☒ urvisha
Frontend	☒ HTML ☒ CSS ☒ JavaScript ☒ React ☒ Vue
Backend	☒ Node.js ☒ Laravel ☒ Django ☒ Spring Boot

Placement Dashboard

127.0.0.1:8000/branches

Placement Dashboard

- Dashboard
- Show Skill
- Reporting
- Manage Skills
- Branch
- Update Profile
- Logout (Placement)

Manage Branches

#	Branch Name
1	CS
2	ME
3	BE
4	BCA
5	MCA
6	Bsc.IT
7	Msc.IT
8	B.Com

Branch Name

Add Branch

32°

Placement Dashboard

127.0.0.1:8000/profile

Placement Dashboard

- Dashboard
- Show Skill
- Reporting
- Manage Skills
- Branch
- Update Profile
- Logout (Placement)

Update Officer Profile

Officer Name

sneha

Email

sneha@student.com

Phone

1234123412

Role

Head

Password (Leave blank to keep current password)

Update Profile

32°

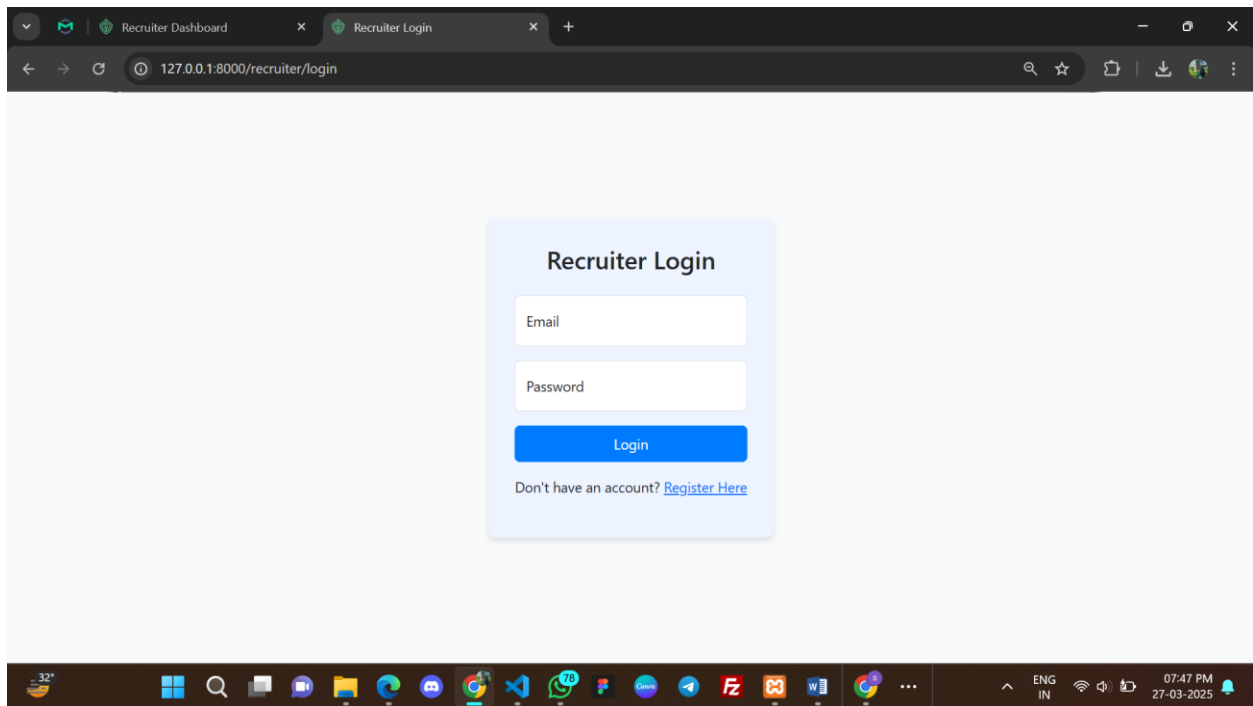
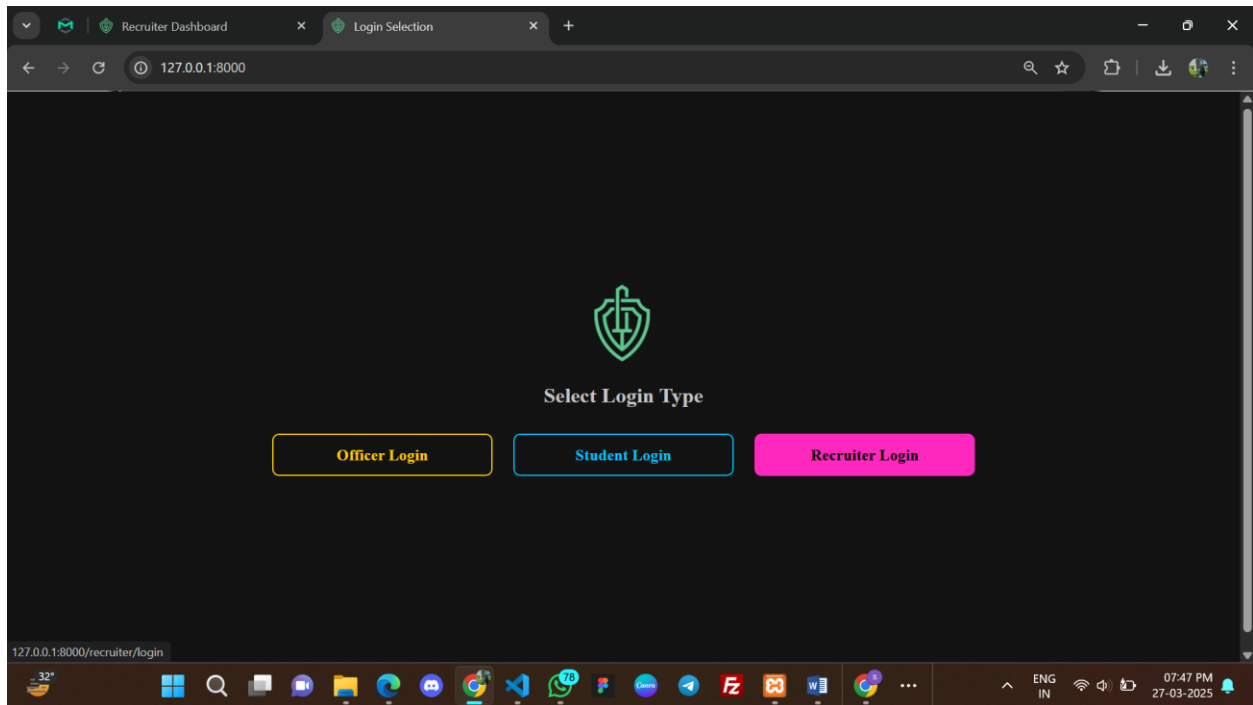
Placement Dashboard

127.0.0.1:8000/profile

Placement Dashboard

- Dashboard
- Show Skill
- Reporting
- Manage Skills
- Branch
- Update Profile
- Logout (Placement)

Recruiter dashboard



Placement Dashboard x Admin x +

127.0.0.1:8000/adminIndex

Welcome, Sneha
Admin

Search

Message v Notification v Sneha v

Dashboard

Placement officers

Students

Branch

Pages v

Logout

Total Students
12

Total Professors
5

Today Branch
8

Total Skills
8

© IntegrateX, All Right Reserved.

↑

Windows taskbar: 32° | Search | File Explorer | Edge | Teams | Chrome | VS Code | WhatsApp | Zoom | Outlook | Word | PowerPoint | OneDrive | Settings | Network | ENG IN | 07:35 PM 27-03-2025

Placement Dashboard x Admin x +

127.0.0.1:8000/placement-officers

Welcome, Guest
Admin

Search

Message v Notification v Sneha v

Dashboard

Placement officers

Students

Branch

Pages v

Logout

Placement Officers

Search Officer by Name

Add Professor

ID	Officer Name	Email	Phone	Role	Action
1	sneha	sneha@student.com	1234123412	Head	Delete
2	urvisha	urvisha@student.com	1231231231	Head	Delete
3	neha	neha@student.com	1231231232	Head	Delete
4	Gopal	gopal@gmail.com	1234567898	Head	Delete
6	Urvisha	urvishalathiya23@gmail.com	98241189229	Head	Delete

Windows taskbar: 32° | Search | File Explorer | Edge | Teams | Chrome | VS Code | WhatsApp | Zoom | Outlook | Word | PowerPoint | OneDrive | Settings | Network | ENG IN | 07:36 PM 27-03-2025

Welcome, Guest
Admin

Dashboard

Placement officers

Students

Branch

Pages

Logout

Search

Students

Search Student by Name

All

ID	First Name	Last Name	Email	Phone	Age	Address	Role	GitHub User ID	Year	Branch ID	Placement Status
1	Sneha	Kalathiya	sneha@gmail.com	1231231231	25	ahmd	Student	snehakalathiya	2024	BCA	Not Placed
2	Rahul	Sharma	rahul@gmail.com	9876543211	24	Ahmedabad,	Developer	urvishalathiya	2025	BCA	Not Placed
3	Priya	Patel	priya@gmail.com	9876543212	23	Mumbai	Designer	dhruvsaidava	2024	ME	Not Placed
4	manav	patel	manav@gmail.com	2222222222	21	gnagar	student	manishamakvana	2025	CS	Not Placed
5	jiya	kalathiya	jiya@gmail.com	3456789778	23	vapi	student	ravisenjaliya	2023	ME	Not Placed
6	manav	makwana	m@gmail.com	9999999999	21	ahmd	student	urvishalathiya	2024	BCA	Not Placed

32°

ENG IN 07:37 PM 27-03-2025

Welcome, Guest
Admin

Dashboard

Placement officers

Students

Branch

Pages

Logout

Search

Manage Branches

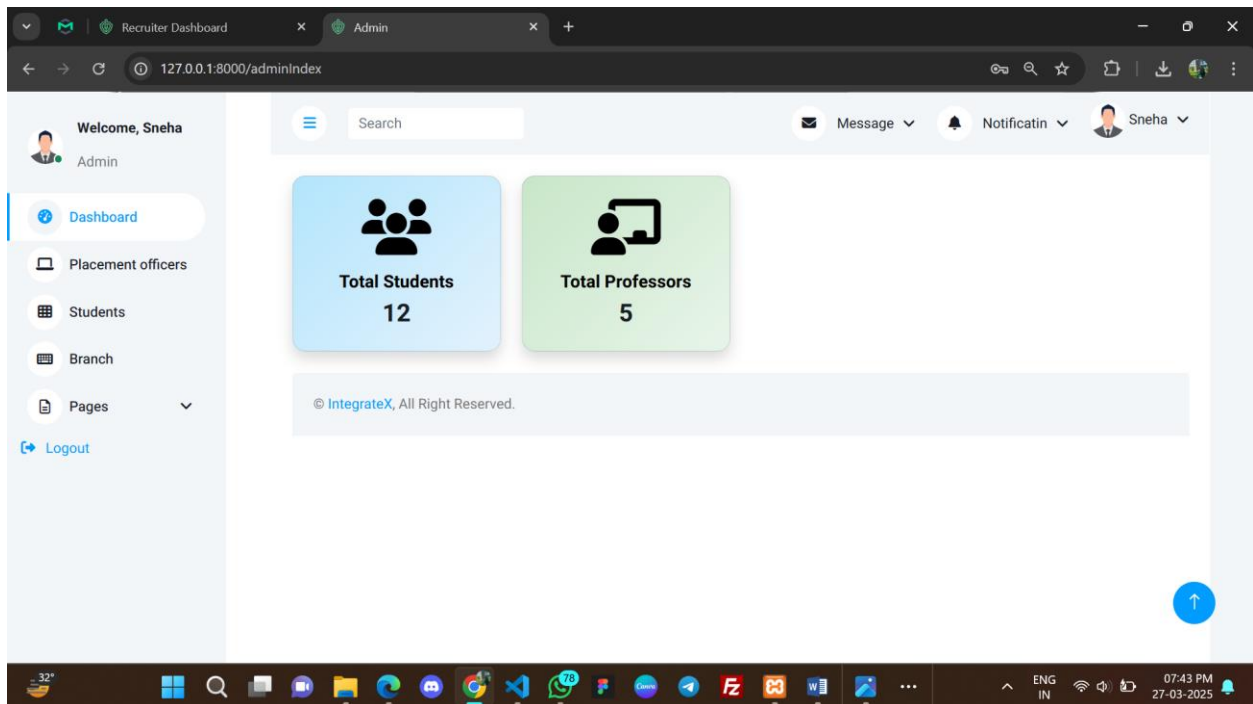
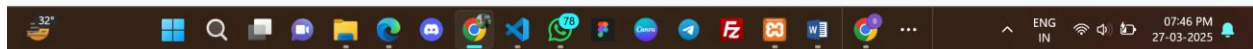
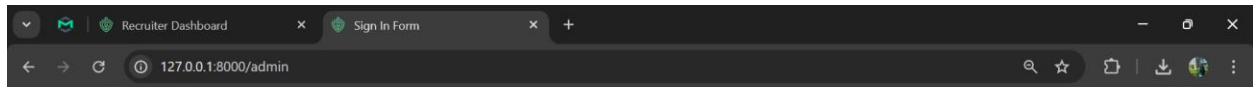
Add Branch

ID	Branch Name
1	CS
2	ME
3	BE
4	BCA
5	MCA
6	Bsc.IT
7	Msc.IT
8	B.Com

32°

ENG IN 07:37 PM 27-03-2025

Admin Dashboard



Recruiter Dashboard

Admin

127.0.0.1:8000/admin/branches

Message

Notificatin

Welcome, Guest

Admin

Dashboard

Placement officers

Students

Branch

Pages

Logout

Manage Branches

Add Branch

ID	Branch Name
1	CS
2	ME
3	BE
4	BCA
5	MCA
6	Bsc.IT
7	Msc.IT
8	B.Com

IntegrateX, All Right Reserved.

32°

Windows Taskbar Icons

ENG IN

07:45 PM

27-03-2025

8. TESTING

Testing is a crucial phase in the development of **IntegrateX**, ensuring the platform operates smoothly, efficiently, and securely.

8.1 White Box Testing

White Box Testing focuses on evaluating the **internal logic, structure, and code** of the platform.

- **Code Coverage Testing:** Ensure every function and condition is tested.
- **Path Testing:** Verify logical paths, including loops and conditional branches.
- **Security Testing:** Identify vulnerabilities, including SQL injection and cross-site scripting (XSS).
- **Laravel Debugging:** Utilize Laravel Telescope to monitor queries and API requests.

8.2 Unit Testing & Integration Testing

Unit Testing ensures that individual components function correctly, while Integration Testing verifies that combined modules work seamlessly.

- **Unit Testing:**
 - Laravel PHPUnit for testing controllers, models, and middleware.
 - Test skill tracking, project uploads, and authentication flows.
- **Integration Testing:**
 - Validate API communication between **frontend (Vue.js) and backend (Laravel 10, MySQL)**.
 - Ensure smooth data retrieval and synchronization across modules.
 - Simulate user interactions such as candidate searches and profile updates.

8.3 Security & Performance Testing

- **Security Testing:**
 - Test **authentication mechanisms** (Laravel Sanctum for API security).
 - Implement role-based access control (RBAC) verification.
 - Scan for **SQL injection, CSRF, and XSS vulnerabilities**.
- **Performance Testing:**
 - Benchmark API response times using **JMeter or LoadRunner**.
 - Evaluate database query performance using **Laravel Query Debugging**.
 - Simulate high traffic loads and concurrent users.

9. STRENGTH OF PROJECT

9.1 Innovation & Impact on Industry

IntegrateX introduces a **validated, structured, and transparent skill assessment system**, solving key inefficiencies in academia and hiring:

- **Skill-Based Hiring Revolution:**
 - Moves beyond traditional resumes with validated **real-world skills**.
- **Automated Matching:**
 - AI-powered candidate filtering and recruiter search tools.
- **Integration with Industry Tools:**
 - Seamless connection with **GitHub, LinkedIn, and certification platforms**.

9.2 Scalability & Future Enhancements

Scalability:

- **Microservices Architecture:** Future transition to microservices for large-scale deployment.
- **Cloud Hosting:** Deployment on **AWS, Google Cloud, or DigitalOcean**.
- **Database Optimization:** Use **MySQL indexing and caching** to enhance speed.

Future Enhancements:

1. **AI-Driven Skill Recommendations:**
 - Personalized career path suggestions based on industry trends.
2. **Blockchain-Based Certification Verification:**
 - Immutable record for skill authentication.
3. **Automated Interview Scheduling:**
 - AI-powered scheduling with integrated video conferencing.
4. **Gamified Learning System:**
 - Encourage student engagement with interactive challenges and leaderboards.

10. FUTURE ENHANCEMENTS

The **IntegrateX Platform** aims to continuously evolve by integrating cutting-edge technologies to enhance **skill validation, recruitment processes, and industry collaboration**. The following future enhancements will help maintain its relevance in a dynamic job market.

- **Advanced Security Measures:** Strengthen authentication with multi-factor authentication (MFA) and end-to-end encryption.
- **Mobile Application Development:** Expand functionality to mobile platforms to increase accessibility.
- **Real-Time Features:** Integrate real-time notifications and live updates using WebSockets.
- **Performance Optimization:** Optimize database queries and implement caching strategies to reduce load times.
 - **Automated Interview Scheduling:**
 - AI-driven scheduling system coordinates recruiter-student availability.
 - Integrates with Google Calendar, Zoom, and Microsoft Teams.

11. USER MANUAL & TECHNICAL DOCUMENTATION

This section provides a **detailed step-by-step guide** for users and administrators to efficiently navigate the **IntegrateX Platform**.

11.1 Step-by-Step Platform Guide

11.1.1 Student Workflow

- 1. Account Registration & Profile Setup:**
 - Register using college email credentials.
 - Complete profile by adding skills, certifications, and project details.
- 2. Skill Tracking & Validation:**
 - Add skills to personal progress tracker.
 - Upload certifications and project links for validation.
- 3. Job Application Process:**
 - Search for job opportunities based on skill match.
 - Apply for jobs directly through the platform.
- 4. Interview & Selection Process:**
 - Attend interviews scheduled through the platform.
 - Receive feedback and track job application status.

11.1.2 Professor & Placement Cell Workflow

- 1. Monitor Student Progress:**
 - Access dashboard for skill checklist tracking.
 - Provide feedback on student progress.
- 2. Generate Placement Reports:**
 - Download analytics on student readiness for recruitment.
 - Share reports with recruiters for candidate shortlisting.
- 3. Update & Customize Skill Checklists:**
 - Modify skill requirements based on industry demands.

11.1.3 Recruiter Workflow

- 1. Candidate Search & Filtering:**
 - Search candidates using specific skill-based filters.
 - View candidate profiles, projects, and certifications.
- 2. Shortlisting & Interview Process:**
 - Shortlist candidates for interview rounds.
 - Schedule interviews through the platform's integrated calendar.
- 3. Final Selection & Hiring:**
 - Extend job offers via the platform.
 - Track candidate onboarding and hiring progress.

11.1.4 Admin Functions

- 1. User Management:**
 - Add, remove, or update user accounts.
 - Assign roles and permissions.
- 2. Skill & Certification Management:**
 - Configure skill categories and industry-aligned validation rules.
- 3. System Analytics & Security Oversight:**
 - Monitor usage analytics and system performance.
 - Ensure compliance with security policies (e.g., GDPR, CCPA)

12.Conclusion

The development of *IntegrateX* has successfully demonstrated a robust Full Stack Development (FSD) framework, integrating frontend, backend, and database systems to provide a seamless user experience. By implementing modern technologies, such as [mention key frameworks used], the project ensures scalability, efficiency, and security. The system effectively meets the intended requirements, offering a user-friendly interface and optimized performance. Through rigorous testing and deployment, the application has proven its reliability and functionality.

13.REFERENCES

<https://getbootstrap.com/>

<https://mailtrap.io/home>

<https://github.com/>

<https://www.linkedin.com/feed/>

<https://themewagon.github.io/skydash/>

<https://lmpixels.com/demo/sunshine-demo/sunshine-version-2/sunshine-html-template/index.html#home>